

Week 1 • Day 3

Pandas Fundamentals & Python–DB Connection

Built the reusable database connection module and mastered core Pandas operations

DATE	INTERN NAME	PROJECT	MENTOR
Day 3 of 28 Week 1		Customer Churn Prediction & LTV Engine	

■ Technologies Used Today

Python

Pandas

SQLAlchemy

NumPy

Jupyter

dotenv

■ Today's Key Metrics

7,043

Rows Loaded from DB

42.7%

M-t-M Churn Rate

32.4mo

Average Tenure

Hidden Empty Strings

■ Day Objectives

- Understand Pandas DataFrame and Series data structures
- Create `src/db/connection.py` — centralised DB connector
- Load dataset from PostgreSQL into a Pandas DataFrame
- Perform basic DataFrame inspection: `shape`, `dtypes`, `head`, `describe`
- Discover churn insights: rate by contract type, tenure statistics

■ Tasks Completed Today

1. Created `src/db/connection.py` with `get_engine()` and `test_connection()` functions
2. Ran `test_connection()` — confirmed connection to PostgreSQL and row count
3. Created `notebooks/explore_basics.py` with full Pandas exploration script

4. Loaded all 7,043 rows from PostgreSQL into df using `pd.read_sql()`
5. Ran `df.shape` → confirmed (7043, 21)
6. Ran `df.dtypes` → identified which columns need type conversion
7. Ran `df.describe()` → found tenure range 0–72 months, avg monthly charge \$64.76
8. Calculated churn distribution: 73.5% No, 26.5% Yes
9. Grouped by Contract to find churn rates: Month-to-month 42.7%, One year 11.3%, Two year 2.8%
10. Calculated tenure statistics: mean 32.4 months, median 29 months
11. Confirmed no NaN values via `df.isnull().sum()` — but TotalCharges has 11 empty strings
12. Installed and launched Jupyter Lab — created notebooks/01_EDA.ipynb
13. Committed all new files to GitHub

■ Concepts Learned

- DataFrame: 2D labelled data structure — rows have index, columns have names
- Series: 1D labelled array — a single column extracted from a DataFrame
- `df.describe()` gives count, mean, std, min, quartiles, max for numeric columns
- `df.groupby()`: how to split data by category and compute aggregates
- `pd.to_numeric(errors='coerce')` converts text to number, invalid values → NaN
- The difference between NaN (Python missing) and empty string ""
- Value counts: `df['col'].value_counts()` vs `normalize=True` for percentages
- Why reading from SQL is better than CSV for reproducibility

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Jupyter Lab	Latest	✓ Installed & running
ipykernel	6.29.0	✓ Installed
Pandas	2.1.4	✓ In use
SQLAlchemy	2.0.23	✓ In use

■ Outputs / Deliverables Produced

- ◆ `src/db/connection.py` — reusable DB connector used by all future modules
- ◆ `notebooks/explore_basics.py` — full data exploration script
- ◆ `notebooks/01_EDA.ipynb` — Jupyter notebook started
- ◆ Key insight printed: month-to-month = 42.7% churn vs two-year = 2.8%
- ◆ Confirmed: 11 rows with empty string in TotalCharges (not true NaN)
- ◆ GitHub commit with `connection.py` and `explore_basics.py`

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
ImportError: cannot import name 'get_engine' from 'src.db.connection'	Added empty src/__init__.py and src/db/__init__.py files — Python needs these to treat folders as packages
KeyError: 'Churn' — column not found	PostgreSQL returns column names in lowercase — used df['churn'] instead of df['Churn']

■ GitHub Commit Record

Commit Message	Week1-Day3: Pandas fundamentals, DB connection module, initial data exploration
Branch	main
Files Changed	src/db/connection.py, notebooks/explore_basics.py, notebooks/01_EDA.ipynb
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ connection.py test passes	■ explore_basics.py runs without errors
■ df.shape = (7043, 21)	■ Churn rate = 26.5% confirmed
■ Contract churn rates printed	■ Tenure stats: mean ~32 months
■ Jupyter Lab opens in browser	■ 01_EDA.ipynb created
■ 11 empty TotalCharges found	■ Day 3 commit on GitHub

■ Self Assessment

8/10 Self Rating	6 Hours Spent	Week 1, Day 3 Progress	3/28 Day in Program
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■ Key Learnings & Reflections

- ★ `src/__init__.py` files are required for Python to treat folders as importable packages
- ★ PostgreSQL column names come back lowercase — always check `df.columns` before filtering
- ★ The 42.7% churn rate for month-to-month contracts is the single biggest business finding so far
- ★ Jupyter notebooks are excellent for exploration but scripts (`.py`) are better for production code

■ Plan for Tomorrow

- Create EDA visualisation charts: churn distribution pie, tenure histogram, monthly charges boxplot
- Build correlation heatmap for numeric features
- Create churn rate bar chart by contract type
- Save all charts to `reports/figures/` folder

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED